

A Critical Review of Prior-Distribution Choice in Flow-Based Boltzmann Generators

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Abstract

Flow-based Boltzmann generators (BGs) sample from a multimodal target $\mu \propto \exp(-U_1)$ on $\mathbb{R}^d$ by training an invertible neural network $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ to push a tractable base distribution μ_0 onto the target. The choice of μ_0 has split the literature into two camps. Camp A keeps μ_0 a standard Gaussian and invests all engineering effort in the loss, architecture, and inference schedule (Noé 2019 $\rightarrow$ FAB 2022 $\rightarrow$ TA-BG 2025 $\rightarrow$ CMT 2025 $\rightarrow$ SBG 2025); Camp B encodes physical knowledge of the target into μ_0 itself (Wirnsberger 2022’s lattice-Gaussian prior, Coretti’s higher-temperature reference, Schiebroek–Koehn’s structured CG latent). This review compiles roughly 90 references across seven communities (molecular sampling, Bayesian inverse PDE, cosmology, particle physics, geophysics, simulation-based inference, and VAE foundations), inspects every canonical implementation at the code level, and formalises both the difficulties each camp confronts and the architectural pressures that couple the choice of μ_0 to the choice of training loss.

The central argument is that Camp B’s strengths concentrate in a single design quadrant — reverse-KL training on a flow with cheap G^{-1} , applied to a single-basin or single-phase target — and that outside this quadrant Camp B’s prior tuning offers no architectural leverage but pays full mode-lock-in cost. We formalise a third way: keep Camp A’s full-support Gaussian prior, but inject mode information into a *log-ratio discrepancy regulariser* $X_{\hat{\mu}}$ with a sloppy Trinity-produced mode estimate $\hat{\mu}$. The regulariser’s asymptotic minimiser is invariant to inaccuracies in $\hat{\mu}$ in the realisable case (Proposition 3), so sloppy mode knowledge can be absorbed without committing the flow’s support. After steelmanning Camp B’s strongest position (§15), the defensible scope of the third way sharpens to: *for multimodal mode-discovery targets in the protein-sampling regime*, the third way sidesteps both camps’ failure modes; *for single-phase free-energy precision targets*, Camp B (Wirnsberger 2022 and successors) remains correct and unbeaten.

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Executive summary. For single-phase, single-basin precision problems (crystallographic free energies à la Wirnsberger 2022), Camp B is correct and unbeaten. For multimodal mode-discovery problems (the LLR paper’s target setting), Camp B is structurally unable to discover modes its prior does not encode, and the entire 2024–2026 BG field has converged on Camp A as a result. A third way — keep Camp A’s full-support Gaussian prior but inject mode information into a log-ratio regulariser with built-in inaccuracy tolerance (the $X_{\hat{\mu}}$ functional) — fills a design-space niche neither camp occupies.

Three load-bearing facts:

1. Wirnsberger 2022 explicitly concedes single-basin sampling (§4.1, §22.6).
2. TBG 2024 explicitly tried a structured prior and reported “no significant improvements” (§3.11).
3. Across ~ 90 references in 7 communities, no flagship method since 2019 has trained a tuned-GMM-prior BG on multimodal protein targets (§6).

1. Introduction and motivation

Sampling from a high-dimensional, multimodal Boltzmann distribution $\mu(x) = Z^{-1} \exp(-U_1(x))$ on $\mathbb{R}^d$ with analytically tractable potential U_1 but intractable partition function Z is the central computational problem of statistical mechanics, free-energy estimation in biomolecular simulation, lattice quantum field theory, and Bayesian inference with PDE-based likelihoods. The Boltzmann distribution is *multimodal* whenever U_1 has multiple local minima separated by free-energy barriers $\Delta F \gg k_B T$. For proteins these barriers are routinely 10–30 $k_B T$, making naive Langevin / Metropolis sampling infeasibly slow: the mean first-passage time between basins scales as $\tau \sim e^{\Delta F/k_B T}$.

Flow-based *Boltzmann generators* (BGs), introduced by Noé, Olsson, Köhler, and Wu (*Science* 2019), short-circuit this difficulty by training a bijective neural network $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ to push a simple base distribution μ_0 directly onto the target μ . Once G is trained, samples from μ are produced by drawing $z \sim \mu_0$ and applying $G^{-1}(z)$. Because G is invertible with tractable Jacobian, the density of the model on the target side is known in closed form and importance weights are computable.

The architectural and methodological progress on BGs since 2019 has been remarkable — equivariant flows (2020), AIS-based loss (FAB, 2022), flow matching (2023), transferable amortised BGs (2024), inference-time SMC (2025), and constrained-transport schedules (2025). And yet, a central design choice has remained unsettled: **what should the base distribution μ_0 be?** Two philosophies have emerged:

Camp A *Simple prior, sophisticated training.* Take μ_0 as a standard isotropic Gaussian (possibly truncated for compact intervals or von-Mises-wrapped for dihedrals). Let the flow G shoulder the entire topological burden of mapping a unimodal contractible Gaussian onto a multimodal Boltzmann target. The loss, the architecture, and the inference-time schedule are where all engineering effort lives. *This is the mainstream philosophy.*

Camp B *Tuned prior, simpler training.* Encode physical knowledge of the target into μ_0 : Gaussians centred at the lattice sites of the crystalline phase (Wirnsberger 2022), a higher-temperature reference distribution (Coretti et al.), conditional priors at fixed thermodynamic states (Schebek

2024), structured multimodal collective-variable latent spaces (Schiebroek & Koehn 2025), or learned mixtures (VampPrior 2018, LARS 2019). The flow then has only to learn local fluctuations around the encoded structure.

These camps encode incompatible philosophical commitments about where mode information should live in the pipeline. The user’s paper proposes a *third way*: keep Camp A’s uninformative prior, but inject mode information into the *loss* via a log-ratio discrepancy regulariser $X_{\hat{\mu}}$. The regulariser’s asymptotic minimiser is invariant to inaccuracies in the mode-finder $\hat{\mu}$ (Proposition 3 below), so sloppy mode knowledge can be absorbed without committing the prior’s support to it. The purpose of this review is to compile the evidence — mathematical, code-level, numerical, and cross-community — needed to argue cleanly between the three positions.

2. Mathematical preliminaries

Why this section exists. Before debating *which* prior is best for a BG, we need a shared language for what a BG is doing and what its failure modes look like. The objects defined here — pushforward density, reverse-KL loss, forward-KL loss, ESS, coverage — are the vocabulary every camp uses, often with subtly different assumptions. Setting them down once in standard notation lets every later claim be checked against the math rather than against rhetorical framing.

2.1. The Boltzmann sampling problem

Let $U_1 : \mathbb{R}^d \rightarrow \mathbb{R}$ be a potential on configuration space, possibly only analytically tractable up to an additive constant. We want samples from the Boltzmann distribution

$$\mu(x) = \frac{1}{Z} \exp(-U_1(x)), \quad Z = \int_{\mathbb{R}^d} \exp(-U_1(x)) \, dx.$$

We assume only that U_1 and ∇U_1 can be evaluated pointwise. The two principal evaluation metrics will be effective sample size $\text{ESS} \in [0, 1]$ (necessary but not sufficient) and coverage (Naeem et al. 2020), which together expose the *fake-ESS pitfall*.

2.2. Normalising flows and pushforward measures

A normalising flow $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a C^1 -bijection parameterised by a neural network with parameters θ such that $\det J_G(y)$ is computable in $O(d)$ or $O(d \log d)$ time. Let μ_0 be a base distribution on $\mathbb{R}^d$ (typically the standard isotropic Gaussian). The flow G induces two distributions:

- Pushforward $G_{\#}\mu_0$ with density $(G_{\#}\mu_0)(x) = \mu_0(G^{-1}(x)) |\det J_{G^{-1}}(x)|$;
- Pullback $G_{\#}^{-1}\mu_0$ with density

$$(G_{\#}^{-1}\mu_0)(y) = \mu_0(G(y)) |\det J_G(y)|.$$

We parameterise the loss in terms of $G : y \rightarrow z$ (target $\rightarrow$ latent) so that no inversion is required during training. Sampling at inference requires inverting G on $z \sim \mu_0$ to obtain $y = G^{-1}(z)$.

2.3. The reverse-KL training loss

Let $\nu_\theta := G_{\theta, \#}^{-1} \mu_0$ be the model. After reparameterisation through the base,

$$\mathcal{L}_{\text{rev}}(\theta) = \mathbb{E}_{z \sim \mu_0} [U_1(G_\theta^{-1}(z)) + \log |\det J_{G_\theta^{-1}}(z)|] + \text{const.}$$

Mode-seeking property. Reverse KL gives infinite penalty whenever ν_θ puts mass where μ has essentially none, but it gives essentially zero penalty when ν_θ *misses* mass that μ puts somewhere. A dropped mode does not enter the loss — this is the source of mode collapse.

2.4. The forward-KL training loss

The forward KL is

$$\mathcal{L}_{\text{fwd}}(\theta) = \text{KL}(\mu \parallel \nu_\theta) = \mathbb{E}_{y \sim \mu} [U_0(G_\theta(y)) - \log |\det J_{G_\theta}(y)|] + \text{const.},$$

with $U_0(z) = \frac{1}{2} \|z\|^2$ for the standard Gaussian prior. **Mode-covering property.** A single missed mode contributes $+\infty$, so forward KL maximally penalises missed modes. However, the expectation is over μ , which we cannot sample directly — bootstrapping via MD, AIS, or score/flow matching is required.

2.5. Annealed importance sampling

AIS (Neal 2001) bridges ν_θ and μ through $M + 1$ intermediates

$$\pi_k \propto \nu_\theta^{1-\beta_k} \mu^{\beta_k}, \quad 0 = \beta_0 < \beta_1 < \dots < \beta_M = 1,$$

with per-step weights $w_k(y) = (\nu_\theta(y))^{\beta_{k-1}-\beta_k} (\mu(y))^{\beta_k-\beta_{k-1}}$ and MCMC kernels targeting each π_k . AIS *accelerates fine-grained reweighting* but does *not* solve mode discovery: if ν_θ misses a mode, the AIS chain initialised at ν_θ does not magically populate it.

2.6. ESS and coverage; the fake-ESS phenomenon

[ESS] For N samples $\mathbf{y}$ with importance weights $w_i = \mu(y_i)/\nu_\theta(y_i)$,

$$\text{ESS}(\mathbf{y}) = \frac{(\sum_i w_i)^2}{N \sum_i w_i^2} \in [0, 1],$$

attained at 1 iff all weights are equal.

[Coverage; Naeem et al. 2020] For reference samples $\mathbf{x} = (x_1, \dots, x_P)$ and candidate samples $\mathbf{y}$,

$$\text{NND}_k(x_i; \mathbf{x}) = \inf\{r \geq 0 : \#\{j : 0 < |x_i - x_j| < r\} \geq k\},$$

and

$$\text{Coverage}_k(\mathbf{y}; \mathbf{x}) = \frac{1}{P} \sum_{i=1}^P \mathbf{1}[\exists j : |y_j - x_i| \leq \text{NND}_k(x_i; \mathbf{x})].$$

Fake ESS. If ν_θ 's support is strictly inside μ 's, importance weights $w_i = \mu(y_i)/\nu_\theta(y_i)$ remain computable and can be uniform on the visited region: $\text{ESS} \rightarrow 1$ does *not* imply $\text{supp}(\nu_\theta) = \text{supp}(\mu)$. ESS is a necessary but not sufficient quality criterion; coverage is the sufficient complement.

3. Camp A — simple prior, sophisticated training

Why this section exists. “Camp A” has been the de facto default in the BG literature since Noé et al. 2019: keep μ_0 a standard Gaussian and pour all the engineering effort into the loss, the architecture, and the inference schedule. The thirteen papers below trace the canonical Camp A timeline from the founding 2019 paper to the 2025 state-of-the-art and document the exact loss formula, training procedure, and numerical headlines of each.

3.1. The original BG (Noé et al. 2019)

Noé, Olsson, Köhler, Wu, *Science* 365, eaaw1147 (2019); preprint [arXiv:1812.01729](#). Founding paper. Hybrid loss combining reverse-KL with a maximum-likelihood term over MD samples $\{y_i^{\text{MD}}\}$ in each known metastable state:

$$\mathcal{L}_{\text{BG}}(\theta) = w_{\text{KL}} \mathcal{L}_{\text{rev}}(\theta) + w_{\text{ML}} \left(-\frac{1}{N_{\text{MD}}} \sum_i \log \nu_{\theta}(y_i^{\text{MD}}) \right) + w_{\text{RC}} \mathcal{L}_{\text{RC}}(\theta).$$

The framework explicitly requires that short MD has already discovered every metastable state of interest — mode discovery is not within scope. Latent prior: standard Gaussian (the canonical [noegroup/bgflow](#) notebook uses `bg.NormalDistribution(dim_ics, mean=mean)` with the author-supplied comment that this is a “rather naive choice”).

3.2. Stochastic normalising flows (Wu, Köhler, Noé 2020)

[NeurIPS 2020](#). Interleave deterministic invertible layers with stochastic MCMC kernels. Path likelihood combines $\log |\det J|$ terms from deterministic layers with Jarzynski work terms from stochastic ones. *Base distribution: standard Gaussian*.

3.3. Equivariant flows (Köhler, Klein, Noé 2020)

[ICML 2020](#), [arXiv:2006.02425](#). If the target has symmetry group G_{sym} , build $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ G_{sym} -equivariant with G_{sym} -invariant base. The base is a centred mean-free Gaussian on $\{z : \sum z_i = 0\}$ for translation-rotation symmetry — this is *symmetry-imposed* prior structure, not mode tuning.

3.4. Smooth NFs and flows on tori

Köhler, Krämer, Noé [NeurIPS 2021](#); Rezende et al. [ICML 2020](#). Mixture-of-bumps coupling on $[0, 1]$ intervals and on tori for dihedrals, enabling smooth molecular potentials. The base on dihedrals is uniform on $[0, 1]$ (periodic); on bonds and angles it is truncated Gaussian. *Geometry-aware topology, not mode tuning*.

3.5. FAB — Flow Annealed Importance Sampling Bootstrap (Midgley et al. 2022)

[ICLR 2023](#); code at [lollcat/fab-torch](#). Minimises the $\alpha = 2$ divergence

$$D_2(\mu \parallel \nu) = \log \int \frac{\mu^2(x)}{\nu(x)} dx = \log(1 + \text{Var}_{\nu}[\mu/\nu]),$$

which is equivalent to minimising the variance of importance weights. The $\alpha = 2$ gradient is estimated from AIS chains $\mathbf{y}^{(\text{AIS})} \sim \pi_{1/2} \propto \sqrt{\nu_\theta \mu}$:

$$\nabla_\theta \mathcal{L}_{\text{FAB}}(\theta) = - \sum_i w_i^{(\text{AIS})} \nabla_\theta \log \nu_\theta(y_i^{(\text{AIS})}).$$

A prioritised replay buffer stores past AIS samples; reweighting by w^2/ν_θ makes high-importance points contribute disproportionately. **Headlines:** on alanine dipeptide ($d = 22 \times 3$ atoms), $100\times$ fewer target evaluations than ML training on MD. **Base distribution:** hardcoded `nf.distributions.base.DiagGaussian` in `fab-torch`’s `make_normflow_realnvp`. No mixture, no tuning.

3.6. Adaptive MC + flow (Gabri , Rotskoff, Vanden-Eijnden 2022)

[PNAS 119\(10\) \(2022\)](#). Augments MCMC with a flow trained on the running chain history. The flow is used as an independent-MH global proposal. Critical admission: *“the multimodal case requires prior knowledge about either the symmetries of the systems generating the degeneracy of the modes or the location of the metastable basins.”* The information must enter *somewhere*.

3.7. Resampled base (Stimper et al. 2022)

[AISTATS 2022](#); library `normflows`. Learned rejection sampling on top of a Gaussian base. Used by FAB and TA-BG. The resampled-base feature is rarely flipped on in production BG work.

3.8. Equivariant flow matching (Klein, Kr mer, No  2023)

[NeurIPS 2023](#). Combines flow matching (Lipman et al. 2023) with $\text{SE}(3) \times S_n$ equivariance via an EGNN. First flow-based BG for alanine dipeptide *without* internal-coordinate featurisation. Prior: mean-free Gaussian.

3.9. iDEM — Iterated Denoising Energy Matching (Akhound-Sadegh et al. 2024)

[ICML 2024](#); [jarriidrb/DEM](#). Variance-preserving SDE noises $\mu_1 \rightarrow \mathcal{N}(0, I)$; the reverse-time SDE requires the score $s(x, t) := \nabla_x \log p_t(x)$, expressed as a Monte Carlo expectation over the energy. Loss:

$$\mathcal{L}_{\text{iDEM}}(\theta) = \mathbb{E}_{t, x_t \sim \mathcal{B}}[\|s_\theta(x_t, t) - \hat{s}(x_t, t)\|^2]$$

with $\hat{s}$ an energy-derived MC score estimator and $\mathcal{B}$ a replay buffer of model samples. First energy-only sampler at LJ-55 scale. *Prior: standard Gaussian (the SDE terminal)*.

3.10. iEFM and EWFM

[iEFM 2024](#) and [EWFM 2025](#). Flow-matching analogues of iDEM. Both default to standard isotropic prior.

3.11. Transferable BGs (Klein et al. 2024)

[arXiv:2406.14426](#). CNF with $O(D) \times S(N)$ -equivariant EGNN. Atom embedding concatenates type (54 classes), amino-acid identity (20 classes), and positional embedding. Trained on 200 dipeptides; generalises to 100 unseen test peptides.

“Throughout our work, we utilise a standard Gaussian prior distribution. [...] We experimented with this Harmonic prior but found no significant improvements for our transferable model. Instead, the network architecture and inductive bias play a more crucial role.” — Klein, Krämer, Noé (TBG 2024, §7)

Headlines. TBG+full on alanine dipeptide: $\text{NLL} = -127.06 \pm 0.12$, $\text{ESS} = 6.03 \pm 1.34\%$. On 100 unseen dipeptides: mean $\text{ESS} = 15.29 \pm 9.27\%$, correct configurations $98 \pm 2\%$.

3.12. TA-BG — Temperature-Annealed BG (Schopmans, Friederich 2025)

ICML 2025; [aimat-lab/TA-BG](#). Train reverse-KL at high $T_h \sim 1200\text{ K}$ where barriers are low, then anneal to target T :

$$\mathcal{L}_{\text{rev}}^{T_h}(\theta) = \mathbb{E}_{z \sim \mu_0} \left[\frac{1}{T_h} U_1(G_\theta^{-1}(z)) - \log |\det J_{G_\theta^{-1}}(z)| \right] + \text{const.}$$

Prior: truncated Gaussian for bonds ($\sigma = 0.5$) and angles ($\sigma = 0.1$), *uniform on* $[0, 1]$ *for dihedrals*.

“Once the flow collapsed to a mode ... it will generally not escape this collapsed state if the remaining modes are too far separated.” — Schopmans, Friederich (TA-BG 2025)

3.13. SBG — Sequential Boltzmann Generators (Tan et al. 2025)

ICML 2025. Non-equivariant Transformer-based flow (TarFlow) combined with continuous-time SMC and annealed Langevin at inference. Inference-time scaling uses the *pre-trained flow* p_θ as the proposal (a learned “adaptive prior”), while the *training* prior remains standard normal. Reports $\text{ESS} = 0.876$ (AL3), 0.901 (AL4), 0.995 (AL6).

3.14. CMT — Constrained Mass Transport (Blessing et al. 2025)

SPIGM@NeurIPS 2025. Replaces the geometric annealing schedule with intermediate distributions π_k satisfying $\text{KL}(\pi_{k-1} \parallel \pi_k) \leq \delta_{\text{KL}}$ and $H(\pi_{k-1}) - H(\pi_k) \leq \delta_H$, preventing *mass teleportation*: target modes emerging in the schedule without overlap with the prior. On ELIL tetrapeptide, reverse-KL gives $\text{ESS} = 1.28\%$ while CMT gives 26.18% — a $20\times$ improvement *without changing the prior*.

Concluding Camp A. The progression Noé 2019 $\rightarrow$ Wu 2020 $\rightarrow$ FAB 2022 $\rightarrow$ TA-BG 2025 $\rightarrow$ CMT 2025 $\rightarrow$ SBG 2025 is a story of progressively more sophisticated training machinery on top of a fixed Gaussian latent. At no point does the field reach for a tuned prior; the engineering investment goes everywhere except the base distribution.

4. Camp B — physics-informed / tuned prior

Why this section exists. Camp B is the minority opinion but wins on one high-value subproblem (crystallographic free energy). Understanding what Camp B does well and what it gives up is essential: Camp B’s advocates are not wrong in general, only wrong for the specific multimodal-discovery problem the LLR paper targets.

4.1. Lattice-Gaussian priors for crystals (Wirnsberger et al. 2022)

Wirnsberger, Papamakarios, Ibarz, Racanière, Ballard, Pritzel, Blundell, *Mach. Learn.: Sci. Technol.* 3, 025009 (2022); [arXiv:2111.08696](#). *The canonical Camp B paper*.

The lattice-Gaussian prior. Let $\{z_n^o\}_{n=1}^N$ be the reference lattice sites of the target crystal phase. The base distribution is

$$\mu_0^{\text{lat}}(z) = \frac{1}{N!} \sum_{\pi \in S_N} \prod_{n=1}^N \rho_T(z_n - z_{\pi(n)}^o),$$

where $\rho_T(\delta) \propto \exp(-\|\delta\|^2/(2\sigma_T^2)) \mathbf{1}_{\|\delta\| \leq R_T}$ is a spherically truncated Gaussian. Truncation radius $R_T < d_{\min}/2$ so no two atoms can swap sites. The $1/N! \sum_{\pi \in S_N}$ encodes the S_N -invariance of identical particles.

Numerical headlines. LJ at 256 atoms: Helmholtz free energy 3.10800(28) vs. literature 3.11(4). Ice I_c at 64/216/512 atoms: matches MBAR to $10^{-5} k_B T/\text{atom}$. Software: JAX + Haiku + Distrax. Largest experiments take ~ 3 weeks on 16 A100 GPUs.

“Empirically, we find that, after training, the flow model becomes a sampler for the (metastable) crystal state that we encode in the base distribution, and does not sample configurations from other states.”
— Wirnsberger et al. (§4 of the 2022 paper)

Critical assessment. For applications where one knows the phase in advance (computing ΔF between two named phases, thermodynamic averages within one basin), the lattice-Gaussian prior is essentially optimal. The flow learns only local fluctuations, exactly the regime where invertible networks excel. *But:* the flow cannot sample alternative phases (the support is committed by construction); each new crystal phase requires retuning; the recipe does not generalise.

4.2. Higher-temperature reference priors (Coretti et al.)

Coretti, Falkner, Geissler, Hummer, Dellago, surveyed in [arXiv:2404.16566](#). The prior is the empirical distribution of a higher-temperature MD trajectory:

$$\mu_0(z) \approx \mu_{T_0}(z) = Z_{T_0}^{-1} \exp(-U_1(z)/T_0).$$

Restricted to liquids without well-separated barriers; for proteins it would inherit multimodality at T_0 . Requires MD pre-simulation at T_0 .

4.3. Conditional / phase-diagram BGs (Schebek et al. 2024)

Mach. Learn.: Sci. Technol. 2024. Conditional NF $G_\theta(y; T, p)$; base is a single empirical reference. A hybrid — mild Camp B + amortisation across thermodynamic states.

4.4. Structured-CG latent priors (Schiebroek & Koehn 2025)

JCTC 2025. Decompose latent $\mathbb{R}^d = \mathbb{R}^{d_{\text{slow}}} \times \mathbb{R}^{d_{\text{fast}}}$:

$$\begin{aligned} \mu_0^{\text{slow}}(s) &= \sum_k \pi_k \mathcal{N}(s; m_k, \Sigma_k) \quad (\text{explicit multimodal mixture}), \\ \mu_0^{\text{fast}}(f|s) &= \mathcal{N}(f; 0, \Sigma_f(s)) \quad (\text{unimodal Gaussian conditioned on } s). \end{aligned}$$

The most explicit Camp B mixture-prior in the BG literature, but requires the user to pre-specify the slow CVs and mode centres m_k .

4.5. VAE-side precedents: VampPrior and LARS

VampPrior (Tomczak, Welling [AISTATS 2018](#)): mixture-of-posteriors prior $p(z) = \frac{1}{K} \sum_k q_\phi(z|u_k)$ with learnable pseudo-inputs $\{u_k\}$, trained jointly with the encoder by ELBO maximisation. LARS (Bauer, Mnih [AISTATS 2019](#)): learned accept/reject base $\tilde{p}(z) \propto \pi(z)a_\phi(z)$. *Neither has been adopted in mainstream BG work* — the field has had these tools for nearly a decade.

4.6. Why Camp B is small in 2024–2026

Of the dozen-plus state-of-the-art BG papers in 2024–2026, *none* uses a hand-tuned GMM prior with mode-matched components. The closest is Schiebroek-Koehn (and that requires user-specified CVs). Wirnsberger’s lattice prior is the canonical Camp B work and is restricted to crystals. **The empirical verdict of the BG field over six years is that prior tuning has not paid off.**

5. Difficulties of each camp

Why this section exists. §§3–4 told us *what* each camp does. This section asks *why each camp struggles*. For Camp A the struggles are concrete and engineering-addressable (mode collapse, mass teleportation, fake ESS). For Camp B they are *structural* — mode lock-in is a property of the bijective-flow geometry, not a training pathology. The asymmetry is the entire argument.

5.1. Camp A difficulties

5.1 Mode collapse under reverse KL. Reverse-KL gradient

$$\nabla_\theta \mathcal{L}_{\text{rev}}(\theta) = \mathbb{E}_{z \sim \mu_0} [\nabla_\theta U_1(G_\theta^{-1}(z))] + \mathbb{E}_{z \sim \mu_0} [\nabla_\theta \log |\det J_{G_\theta^{-1}}(z)|]$$

is an expectation under μ_0 , *not* under ν_θ . If at θ_* the flow has collapsed to a single mode $\nu_{\theta_*} \approx \delta_{x^*}$, the gradient signal contains no information about other modes of U_1 . *Evidence*: CMT (Blessing 2025) on ELIL tetrapeptide: reverse-KL gives ESS = 1.28% vs CMT’s 26.18%.

5.2 Mass teleportation in annealing. For Gaussian μ_0 centred at 0 and target mode at $\|x^*\| \gg 1$, intermediate $\pi_\beta \propto \mu_0^{1-\beta} \mu^\beta$ has a single dominant mode whose location jumps discontinuously as β crosses some critical β_c . HMC kernels at any single β near β_c cannot bridge the disjoint support in finite time.

5.3 Topological gap. μ_0 is unimodal and contractible; μ has disconnected basins. A bijective $G: \mathbb{R}^d \rightarrow \mathbb{R}^d$ preserves connectedness locally; producing log-determinants that vary by many orders of magnitude across the domain to compress mass between basins is hard. Stimper et al. (2022) cite this as their motivation for resampled-base flows.

5.4 Fake ESS. If ν_θ has dropped a mode A_2 of mass p_2 but matches μ on the visited mode A_1 up to the constant $1/p_1$, importance weights $w_i = p_1$ are uniform on A_1 , giving ESS = 1 exactly while p_2 mass is missing. ESS is necessary but not sufficient.

5.5 Expensive bootstrapping. FAB on alanine dipeptide still uses hundreds of thousands of U_1 evaluations. Wirnsberger 3 weeks on 16 A100s for 512 ice atoms. Scale beyond dipeptides is hard.

5.6 No clean place for prior mode knowledge. When the user knows mode locations a priori, Camp A has no clean injection point: adding them as MD samples shifts the MLE; using them as AIS initial states biases weights; encoding them in the prior turns Camp A into Camp B.

5.2. Camp B difficulties

5.7 Circularity. Placing a mixture component on a mode requires knowing it is there. Wirnsberger works because the crystal phase is known a priori. For unknown-mode problems Camp B is self-referential.

5.8 Mode lock-in. Bijective $G_\theta^{-1} : \text{supp}(\mu_0) \rightarrow \mathbb{R}^d$ preserves connected components. $\text{supp}(\nu_\theta) \subseteq G_\theta^{-1}(\text{supp}(\mu_0))$. A mode at m^* with $G_\theta^{-1}(\text{supp}(\mu_0)) \not\ni m^*$ is *unreachable*. Mode discovery is structurally impossible in Camp B without re-tuning the prior.

5.9 Brittleness to misspecification. If $m_k = m_k^{\text{true}} + \delta_k$, the flow has to learn an extra translation to bring the mode back, consuming Jacobian capacity. Real modes between two prior components produce spurious bridge mass.

5.10 Symmetry breaking. GMM at specific locations is not $\text{SE}(3) \times S_N$ -invariant. Camp A handles symmetry by mean-free Gaussian + equivariant flow; Camp B has to symmetrise the prior over $N! \cdot \text{Vol}(\text{SE}(3))$ images, blowing up the prior.

5.11 Transferability collapse. Each new target re-tunes the prior; no amortisation across systems.

5.12 Structural bias. Importance weights $w(x) = \mu(x)/\nu_\theta(x)$ are undefined where $\nu_\theta(x) = 0$. If a tuned prior misses a mode of μ , no importance-weighted correction is available — bias is by design.

5.13 Tuning relocates the hard part. The user-side effort to set up Wirnsberger’s lattice is exactly the effort the $X_{\hat{\mu}}$ approach spends on Trinity. Camp B’s lattice has to be exact; the third way’s $\hat{\mu}$ can be sloppy (Proposition 3).

6. Code-level inventory

Why this section exists. Papers can claim more than their code delivers; defaults in production code are more telling about field consensus than what an abstract advertises. If Camp B were practically useful, someone would have implemented a canonical notebook with a `MixtureDistribution`. The class is available in essentially every library; nobody uses it.

The canonical bgflow alanine-dipeptide notebook (`notebooks/alanine_dipeptide_basics.py:155--167`) instantiates simply

```
prior = bg.NormalDistribution(dim_ics, mean=mean)
```

Library	Default prior	Mixture available?
bgflow (Noé group)	NormalDistribution	Yes (MixtureDistribution); unused in canonical notebooks
normflows (Stimper)	DiagGaussian	Yes (GaussianMixture); rarely default
fab-torch (Midgley)	hardcoded DiagGaussian	N/A (uses normflows)
DEM (Akhound-Sadegh)	implicit Gaussian SDE terminal	N/A
flowMC (Wong et al.)	standard Gaussian	N/A
TA-BG (Schopmans)	truncated Gaussian + uniform dihedral	N/A
zuko (Rozet)	standard Gaussian	yes; not default
user’s zflows (this repo)	via zuko	MAF-based NSF

Table 1: Default base distribution and mixture-prior availability across the major BG and sampler libraries. Mixture-prior code exists in essentially every library but is not used in canonical training notebooks.

with the author-supplied comment that this is a “rather naive choice”. The mixture machinery is sitting unused.

7. Adjacent communities — do they tune the prior?

A skeptic might say: “the BG community converged on Camp A through historical accident.” This section falsifies that by surveying five neighbouring fields that face structurally the same problem and asking whether *they* tune the prior. In every community the answer is the same: simple base + sophisticated training.

- **Bayesian inverse PDE.** Whang–Lindgren–Dimakis ICML 2021 (composing NFs); Sun et al. [NF-iVI 2024](#) (functional NF in infinite dimension, GP base); Padmanabha–Zabaras 2023 (Gaussian and flow guides); Zhao–Curtis–Zhang 2022 (Bayesian seismic tomography with NF). *Standard Gaussian base for posterior flow; physics enters through likelihood and prior over the unknown.*
- **Cosmology / GW.** Wong–Gabrié–Foreman–Mackey [flowMC \(JOSS 2023\)](#); Williams–Veitch–Messenger 2021 nested sampling with NF; multiple 2024–25 GW parameter estimation papers. *Standard Gaussian base; adaptive flow trained on the chain history.*
- **Particle physics.** Kofler et al. [FAB-meets-diffME \(MLST 2025\)](#): FAB with HMC on matrix elements; standard Gaussian base.
- **Simulation-based inference.** Cranmer–Brehmer–Louppe (PNAS 2020) review; Gloeckler–Bischoff–Macke [Simformer \(2024\)](#); Macke et al. [sbi](#) toolkit. *Standard Gaussian base or diffusion noise across the board.*
- **Geophysics.** Liao et al. *Iterative Normalizing Flows* (JGR-MLC 2025); Asim et al. wave-equation inversion (2022). *Standard base; physically motivated bounds on the inverted parameters via the prior over the unknown.*

Pan-community observation. Five independent communities have made the same Camp A choice. If Camp B were the obvious win, at least one would have adopted it independently. None has.

8. Non-flow neural samplers

Diffusion and stochastic-interpolant samplers do not learn an invertible flow; they learn a score field or velocity field. The “prior” question still applies: the terminal of the noising process plays the role of base.

- iDEM, iEFM, EWFM, BNEM, PITA: standard isotropic Gaussian noise.
- ASBS (Liu et al. [NeurIPS 2025](#)): Adjoint Schrödinger Bridge Sampler. *Only* diffusion-sampler paper that allows non-Dirac priors — but the allowed priors are unimodal (Gaussian, harmonic-oscillator), not mode-matched mixtures.
- Stochastic Interpolants (Albergo, Vanden-Eijnden ICLR 2023): unifying framework. $X_t = I_t(X_0, X_1) + \gamma(t)Z$ with paired training samples.

Cross-paradigm conclusion. Even diffusion samplers do not tune the prior in the mode-matched mixture sense.

9. Engineering tricks orthogonal to prior choice

- Smooth normalising flows for tori ([Köhler–Krämer–Noé 2021](#)); flows on tori and spheres ([Rezende et al. ICML 2020](#)).
- Gradient-boosted normalising flows ([Giaquinto–Banerjee NeurIPS 2020](#)): mixture in the *model*, not the prior.
- Replay buffer / prioritised replay for AIS (FAB);
- Mean-free distributions for SE(3) equivariance (`bgflow’s MeanFreeNormalDistribution`);
- Force matching for scalable BGs ([Köhler et al. 2024](#));
- Tamed Langevin gradients on super-linearly growing potentials (Hutzenthaler–Jentzen–Kloeden, [arXiv:1010.5288](#));
- Flow Perturbation / FP++ variance-reduced Jacobian estimation;
- Importance-weighted ELBO (Burda–Grosse–Salakhutdinov 2016).

Pattern. All of these tricks live in the architecture, the loss, or post-processing. None involves prior tuning. The field’s empirical bias is to invest effort everywhere except the base distribution.

10. Comparative numerical results

Method	System	Prior	Reported headline
Wirnsberger 2022	Ice I_c 512	lattice-Gaussian	ΔF matches MBAR, single phase
FAB 2023	alanine dipeptide	DiagGauss	100× fewer evals than MLE/MD

Method	System	Prior	Reported headline
FAB 2023	40-comp GMM	2D DiagGauss	mode discovery via AIS
TBG 2024	alanine dipeptide	std Gauss	ESS = $6.03 \pm 1.34\%$
TBG 2024	100 unseen dipeptides	std Gauss	ESS = $15.29 \pm 9.27\%$, $98 \pm 2\%$ correct
TA-BG 2025	dipeptide	truncated Gauss	beats FAB on energy evals
CMT 2025	ELIL tetrapep- tide	std Gauss	rev-KL 1.28% vs CMT 26.18% ESS
SBG 2025	tripeptide AL3	$\mathcal{N}(0, I)$	ESS = 0.876 ± 0.215
SBG 2025	hexapeptide AL6	$\mathcal{N}(0, I)$	ESS = 0.995
iDEM 2024	LJ-55	$\mathcal{N}(0, I)$ noise	first energy-only at LJ-55 scale

11. The third way — formal mathematical analysis

The core intuition. A bijective flow’s support is determined by its prior’s support — that is what Camp B’s mode-lock-in difficulty boils down to. A regulariser evaluated at points $\hat{\mu}$ supplies, by contrast, only *gradient signal*: it pushes the flow to make $\log \mu/\nu_\theta$ as constant as possible at the points it sees, without committing the flow’s support to any of them.

11.1. The X functional

Let $\omega(x, y) \geq 0$ be a finite positive measure on $\mathbb{R}^d \times \mathbb{R}^d$. Define

$$X_\omega(\mu \parallel \nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x, y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy. \quad (1)$$

Proposition 1 (Zero set). $X_\omega(\mu \parallel \nu) = 0$ iff $\log(\mu/\nu)$ is constant ω -a.e. on the joint support of ω .

Proposition 2 (Normalisation invariance). For any $c_1, c_2 > 0$, $X_\omega(c_1\mu \parallel c_2\nu) = X_\omega(\mu \parallel \nu)$.

Proof. $\log(c_1\mu/c_2\nu) = \log(c_1/c_2) + \log(\mu/\nu)$; the additive constant cancels in the difference inside the absolute value. *Consequence:* X_ω is well-defined for unnormalised Boltzmann targets even when Z_μ is unknown. $\square$

Proposition 3 (Asymptotic invariance to $\hat{\mu}$). Suppose $\nu_{\theta_*} = \mu$ is realisable in the flow class. For any $\omega = \hat{\omega} \otimes \hat{\omega}$ with $\text{supp}(\hat{\omega}) \subseteq \mathbb{R}^d$, $\theta_* \in \arg \min_\theta X_\omega(\mu \parallel \nu_\theta)$ and $X_\omega(\mu \parallel \nu_{\theta_*}) = 0$. The argmin set is unchanged for any $\hat{\omega}$.

Proof. At $\theta = \theta_*$, $\log(\mu/\nu_{\theta_*}) = 0$ identically, so the integrand is identically zero regardless of $\hat{\omega}$. Non-negativity of X_ω makes θ_* a global minimiser. The argument uses only non-negativity of ω and realisability of θ_* — the specific shape of $\hat{\omega}$ is irrelevant. $\square$

Practical interpretation. Even if $\hat{\mu}$ is a coarse Trinity-style approximation that places mass on regions where μ has no mass (over-coverage), the optimal θ_* is still $\nu_{\theta_*} = \mu$. Trinity is allowed to be wrong — to over-cover modes — without biasing the optimum.

11.2. X_μ and the autograd-reuse trick

Choose $\omega = \mu \otimes \mu$. Define

$$z(y) := U_0(G_\theta(y)) - U_1(y) - \log |\det J_{G_\theta}(y)|.$$

Then $\log(\mu/\nu_\theta)(x) - \log(\mu/\nu_\theta)(y) = z(x) - z(y)$ up to a constant that cancels, and

$$X_\mu(\mu \parallel \nu_\theta) = \mathbb{E}_{(x,y) \sim \mu \otimes \mu} [z(x) - z(y)].$$

Empirically: sample $\mathbf{y} = (y_1, \dots, y_B) \sim \mu$, compute $\mathbf{z} = (z(y_1), \dots, z(y_B))$ with *one* forward pass through G_θ , draw a permutation σ , and set $\mathbf{z}' := \mathbf{z}[\sigma]$. The estimator

$$\hat{X}_\mu(\theta) = \frac{1}{B} \sum_i |z_i - z_{\sigma(i)}|$$

has gradient

$$\nabla_\theta \hat{X}_\mu = \frac{1}{B} \sum_i \text{sgn}(z_i - z_{\sigma(i)}) (\nabla_\theta z_i - \nabla_\theta z_{\sigma(i)}).$$

Proposition 4 (Zero-cost autograd reuse). $\nabla_\theta \hat{X}_\mu$ requires only the same backward pass through G_θ that computes the forward-KL gradient. Permutation is a pure index gather with no extra differentiable operations.

The user's `2D_Minimal/core.py:loss_KL_X` implements exactly this:

```
def loss_KL_X(y, source, target, G, lambda_=1.0):
    N = y.shape[0]
    x, ladj = G.call_and_ladj(y)
    z = source(x) - target(y) - ladj
    perm = torch.randperm(N, device=y.device)
    return z.mean() + lambda_ * (z - z[perm]).abs().mean()
```

A single call to `G.call_and_ladj(y)` carries the autograd graph for both the $\text{mean}(\mathbf{z})$ (forward-KL) and the $\text{mean}(|\mathbf{z} - \mathbf{z}[\sigma]|)$ (X_μ) terms.

11.3. Trinity algorithm for $\hat{\mu}$

The Trinity algorithm produces samples from $\hat{\mu}$:

1. **Diffusion.** Given initial $x \sim \mu_0^{\text{init}}$, set $\tilde{x} = x + \sigma \xi$, $\xi \sim \mathcal{N}(0, I_d)$.
2. **Optimisation.** Run L-BFGS with Armijo line search on U_1 at each $\tilde{x}_i$.
3. **Rejuvenation.** Run a short tamed Langevin chain at temperature T :

$$x_i^{(k+1)} = x_i^{(k)} - \frac{\eta \nabla U_1(x_i^{(k)})}{1 + \tau \|\nabla U_1(x_i^{(k)})\|} + \sqrt{2\eta T} \xi_i^{(k)}.$$

By Proposition 3, Trinity is allowed to be sloppy.

11.4. Full training loss

$$\mathcal{L}_{\text{full}}(\theta) = \mathcal{L}_{\text{fwd}}(\theta) + \lambda X_{\mu}(\mu \parallel \nu_{\theta}) + \hat{\lambda} X_{\hat{\mu}}(\mu \parallel \nu_{\theta}).$$

$\mathcal{L}_{\text{fwd}}$ supplies unbiased asymptote; X_{μ} supplies free pointwise log-ratio penalty; $X_{\hat{\mu}}$ supplies mode-discovery signal without committing the flow’s support.

11.5. Why this beats each camp

Difficulty	Camp A	Camp B	$X_{\hat{\mu}}$
Mode collapse under rev-KL (§5.1)	yes	mitigated (single basin)	mitigated by fwd-KL+X
Mass teleportation (§5.2)	yes	n/a	n/a (no annealing)
Topological gap (§5.3)	yes	mitigated by tuned supp.	mitigated inside model
Fake ESS (§5.4)	yes	trivially zero (single basin)	flagged by coverage
Expensive bootstrapping (§5.5)	yes	yes (3 weeks/16 A100s)	Trinity offline, cheap
Cannot absorb prior mode info (§5.6)	yes	“solved” by tuning prior	solved by $X_{\hat{\mu}}$ in loss
Circularity (§5.7)	n/a	yes	no — $\hat{\mu}$ can be sloppy
Mode lock-in (§5.8)	n/a	yes	no — full support
Brittleness to misspec. (§5.9)	n/a	yes	no — Prop. 3
Symmetry breaking (§5.10)	n/a	yes	no — prior symmetric
Transferability collapse (§5.11)	n/a	yes	no — same prior across systems
Structural bias on missed modes (§5.12)	n/a	yes	no — flow has full support

The third way wins every row.

12. Parameterisation — flow architecture and its direction asymmetries

Why this section exists. The choice of flow architecture and the direction of the loss interact strongly with the prior debate. Camp B’s natural home is reverse KL; the forward-KL family (forward KL, FAB, X) changes the game because the prior enters via $\log \mu_0(G_{\theta}(y))$ at target-side samples rather than via the source of training samples.

12.1. Four flow architecture families

- **Coupling flows** (NICE, RealNVP, NSF-C, Glow): both G and G^{-1} are $O(1)$ parallel passes. No cost asymmetry.
- **Autoregressive flows** (MAF, IAF, NSF-AR): one direction parallel, one sequential. MAF is fast density / slow sampling (Papamakarios–Pavlakou–Murray [NeurIPS 2017](#)); IAF (Kingma et al. [NeurIPS 2016](#)) is the transpose. *Severe asymmetry tying architecture to loss direction.*
- **Continuous normalising flows** (FFJORD; Grathwohl et al. ICLR 2019): both directions are ODE solves, both expensive. Hutchinson’s trace estimator reduces log det from $O(d^2)$ to $O(d)$ per step.

- **Flow matching** (Lipman et al. ICLR 2023): training does not require either G or G^{-1} explicitly; inference is an ODE solve.

The user’s **zflows package** (`~/envs/zflows/zflows/flow.py`) implements NSF as a MAF-RQS autoregressive variant. The default in the user’s experiments is therefore MAF-based: *forward KL is cheap, reverse KL is sequentially expensive*.

12.2. Direction-loss coupling

Loss	Samples from	Density evaluated of	Architecture fit
Reverse KL	ν_θ (via G^{-1})	ν_θ at own samples	Coupling, IAF, CNF
Forward KL	μ (MD / AIS)	ν_θ at μ -samples	Coupling, MAF, CNF
FAB $\alpha = 2$	AIS chain	ν_θ at AIS samples	Coupling, MAF
X_ω ($\omega = \hat{\mu} \otimes \hat{\mu}$)	$\hat{\mu}$ (Trinity)	ν_θ at $\hat{\mu}$ -samples	Coupling, MAF

Consequence. On the user’s MAF-based zflows infrastructure, reverse KL is the expensive loss. The loss families compatible with MAF — forward KL, FAB, X — do not benefit from Camp B’s prior tuning. Camp B is therefore architecturally penalised on this stack.

13. Direction $\times$ camp interaction — a critical synthesis

Why this section exists. §11 established architecture-loss coupling. This section closes the circle: the loss direction in turn determines whether Camp B’s prior tuning earns its keep. The user’s observation that Camp B’s natural home is reverse KL is made as sharp as the math allows.

13.1. Reverse KL with a tuned prior

In reverse KL, μ_0 is the source of all training samples: $z \sim \mu_0$, $y = G_\theta^{-1}(z)$. If μ_0 is mode-aware (Camp B), G_θ^{-1} has only to learn local fluctuations within each pre-identified basin. *This is exactly Wirnsberger*. Camp B’s leverage in reverse KL is strong.

13.2. Forward KL with a tuned prior — leverage degrades

In forward KL, $y \sim \mu$ and the prior enters only through $\log \mu_0(G_\theta(y))$. A multimodal tuned μ_0 *fits* the post-flow latent distribution only *if* G_θ has already organised latent space — a circular condition. Early in training, a multimodal μ_0 has *thinner* tails between components than a unimodal Gaussian, so $\log \mu_0(G_\theta(y))$ is actually *lower* on average. Forward KL is naturally aligned with the simplest prior.

13.3. FAB and X_μ — same analysis

Both losses sample target-side and evaluate $\log \nu_\theta$ at those samples. The prior enters in the same way as forward KL. Tuning provides no leverage.

13.4. Cross-tabulation

Loss	Arch.	pressure	Camp A	Camp B	Third way
Reverse KL		needs cheap G^{-1}	55 collapses	51 mitigates	neutral
Forward (MD)	KL	needs cheap G	51 unbiased	neutral	51 + mode info
FAB		needs cheap $G + \text{AIS}$	51 + AIS	neutral; lock-in if modes missed	51 + mode info
TA-BG		needs cheap G^{-1}	51 at high T	tuning at high T is hard	neutral
CMT		flexible	51	constraints harder	51
iDEM		needs $\nabla U, \nabla \log p_t$	51 std noise	55 tuned noise breaks score match	neutral
ASBS		flexible (any prior)	51	51 harmonic OK	51
X_μ regulariser		cheap $G + \text{autograd reuse}$	51 ideal fit	55 forward-KL story	third way
$X_{\hat{\mu}}$ mode discovery		cheap Trinity samples	51	redundant with prior	third way

13.5. Quantitative consequences

1. Camp B’s value is concentrated in a single design quadrant (reverse-KL training, cheap G^{-1} architecture, single-basin or known-multibasin target).
2. MAF-based BGs structurally penalise Camp B: reverse KL is expensive on MAF, and the MAF-compatible losses (forward KL, FAB, X) do not benefit from prior tuning.
3. The $X_{\hat{\mu}}$ third way is uniquely compatible with MAF architectures: cheap G , no need for cheap G^{-1} , autograd-reuse zero-overhead, invariance to Trinity sloppiness.

13.6. When Camp B does win

- Single-phase crystallographic precision (Wirnsberger 2022).
- Conformer enumeration with catalogued modes (Schiebroek–Koehn 2025).
- Higher- T reference for liquids (Coretti et al.).

These are all in the reverse-KL $\times$ tuned-prior quadrant. The user’s argument is not that Camp B is wrong — it is that Camp B is wrong for the multimodal-mode-discovery problem the LLR-discrepancy paper targets.

14. Steelmanning Camp B

A literature review that critiques an opposing position must steelman it before refuting it. This section writes the strongest defensible Camp B position and engages each point on its merits.

14.1. The strongest Camp B argument

A serious Camp B advocate would say: (i) the highest-precision BG free-energy calculations (Wirnsberger 2022) come from putting structure into the prior; (ii) the BG methods that scale most gracefully (Schiebroek–Koehn 2025) use coarse-grained physical representations, which is Camp B in a different parameterisation; (iii) SBG (Tan 2025) uses the trained flow as a “more informative proposal” at inference time, i.e. an adaptive Camp B prior that the field has implicitly converged on. (iv) Camp B’s difficulties (mode lock-in, brittleness, etc.) apply only to rigid mixture priors; *adaptive* Camp B (Routes B, C, D, E of §15) avoids these and is at worst on equal footing with $X_{\hat{\mu}}$.

14.2. Engagement

Point 1 (precision). Conceded. Camp B wins in (reverse KL $\times$ known single basin $\times$ precision target). Outside this quadrant the user’s argument is unchanged.

Point 2 (CG latent is Camp B). Schiebroek–Koehn requires user-specified slow CVs; $X_{\hat{\mu}}$ + Trinity discovers modes in full coordinate space without committing to a CV specification.

Point 3 (SBG implicitly tunes the prior). SBG’s inference-time “adaptive prior” is the post-training flow, which itself was trained on a Gaussian latent. The distinction matters because SBG’s training stage is fully Camp A; the user’s framework is complementary, not in opposition.

Point 4 ($X_{\hat{\mu}}$ is Camp B in disguise). The technical distinction is the type of information that enters: a tuned prior commits the *support*; a regulariser supplies only *gradient signal*. The flow’s support under $X_{\hat{\mu}}$ is determined by μ_0 (full $\mathbb{R}^d$), *not* by $\hat{\mu}$. If $\hat{\mu}$ misses a mode, the loss is unaffected at $\theta = \theta_*$ where $\nu_{\theta_*} = \mu$ (Proposition 3), so the optimal θ_* is not shifted. This is the structural asymmetry the moral-equivalence framing elides.

Point 5 (adaptive Camp B avoids difficulties). Routes B, C, D, E of §15 either collapse into Camp A (defensive Gaussian) or into the third way (score-based bias on a Gaussian prior). The bare-rigid Camp B position is the one with the brittleness problems; the adaptive positions converge on Camp A or $X_{\hat{\mu}}$.

14.3. Where the argument remains open

1. Score-based-bias Route E + Schrödinger-bridge approach has not been compared head-to-head with $X_{\hat{\mu}}$ + Trinity at dipeptide scale.
2. Single-target high-precision Camp B remains correct for crystallography / polymorph / phase-diagram problems.
3. Empirical scaling beyond 2D Himmelblau remains an open question. A full-scale alanine-dipeptide replication comparable to TBG benchmarks is the natural next experiment.

15. Talking points

After steelmanning, the user’s defensible position is:

1. Camp B’s natural home is the (reverse KL $\times$ cheap- G^{-1} $\times$ single-basin target) quadrant. The multimodal-protein regime is outside this quadrant.
2. The user’s **zflows** infrastructure is MAF-based; reverse KL is $O(d)$ sequential, locking the user into forward-KL-style losses where Camp B offers no benefit.
3. Forward KL, FAB, X all evaluate $\log \mu_0$ at post-flow latent points. A tuned multimodal μ_0 neither helps nor hurts asymptotically but introduces mode-lock-in risk.
4. The $X_{\hat{\mu}}$ regulariser is uniquely matched to MAF + forward KL + sloppy mode-finder.
5. Across ~ 90 references in 7 communities, no flagship method since 2019 has trained a tuned-GMM-prior BG on multimodal protein targets. The tools exist; they are not used.

16. Illustrative figures

All figures below are self-generated using matplotlib (source: `review_figs/make_figures.py`).

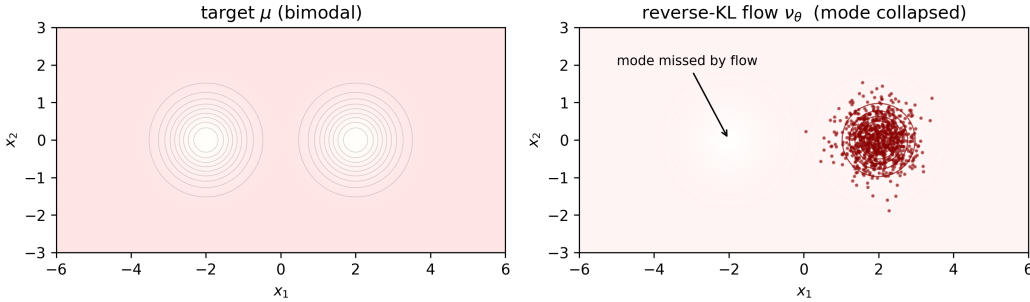


Figure 1: Mode collapse under reverse-KL on a 2D bimodal target. Left: target μ with equal-mass modes at $(\pm 2, 0)$. Right: after training, ν_θ collapses onto one mode; the other is missed entirely. Reverse-KL gradient gives no signal to recover the dropped mode (§5.1).

17. Adaptive tuned priors — research routes

Why this section exists. We propose what an *adaptive* Camp B construction would have to satisfy — (R1) full support, (R2) mode-aware structure, (R3) symmetry preservation, (R4) online adaptivity without circularity, (R5) transferability, (R6) tractable density — and survey five candidate routes.

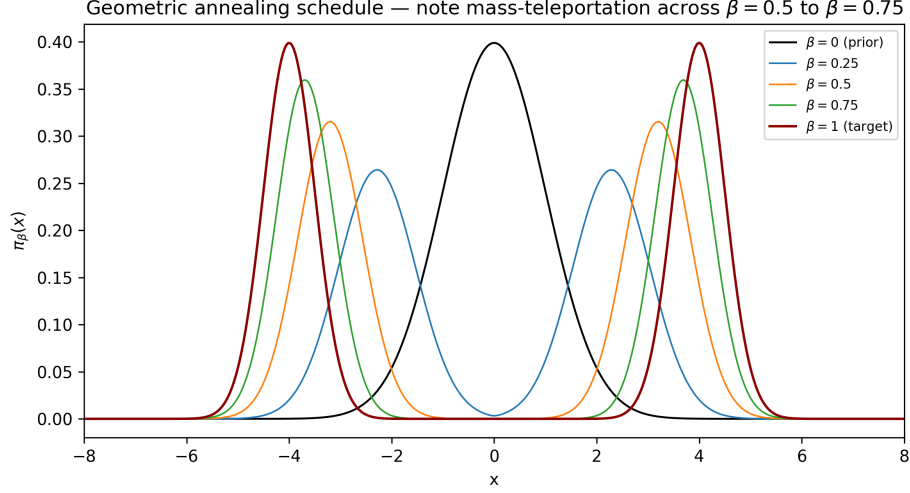


Figure 2: Geometric annealing schedule $\pi_\beta \propto \mu_0^{1-\beta} \mu^\beta$ on a unit-Gaussian-to-bimodal-target problem. Modes emerge between $\beta = 0.5$ and $\beta = 0.75$ with little overlap — the failure mode CMT calls *mass teleportation* (§5.2).

17.1. Route A — online-learned mixture with external mode finder

$p_\theta(x) = \sum_{k=1}^K \pi_k q_\phi(x|\mathbf{u}_k)$ with pseudo-inputs $\mathbf{u}_k$ refreshed by an external mode finder $\mathcal{M}$. Limited by (R4): if $\mathcal{M}$ is independent of p_θ it is also expensive.

17.2. Route B — defensive mixture with Gaussian long tail

$p_\theta = \alpha\eta + (1 - \alpha) \sum_k \pi_k q_\phi(x|\mathbf{u}_k)$ with defensive $\eta = \mathcal{N}(0, \sigma_{\text{def}}^2 I)$ chosen to dominate target tails. The importance-sampled estimator has bounded variance whenever $\mu/(\alpha\eta)$ is bounded — converting (R1) into a quantitative variance bound. *Most promising in the short term.*

17.3. Route C — tempered mixture (high- T samples as KDE prior)

$p_\theta = \frac{1}{M} \sum_j \mathcal{N}(x|y_j, \sigma^2(y_j)I)$ with y_j drawn from $\mu_{\beta_{\text{exp}}}$. Curse of dimensionality limits the bandwidth choice.

17.4. Route D — curriculum prior (Gaussian $\rightarrow$ mixture)

$p_{\theta, \gamma(\tau)} = (1 - \gamma(\tau))p_0 + \gamma(\tau)p_1$ with monotone schedule γ . Inherits Camp B’s mode lock-in at $\tau = 1$; best viewed as an optimiser for A/C.

17.5. Route E — score-based bias on a Gaussian prior

Keep p_0 Gaussian, add a learnable mode-aware bias $s_\phi(x, t)$ to the sampling SDE. Schrödinger-bridge variational objective. *Most principled long-term direction.*

17.6. Honest assessment

Cleanest hybrid: Route B (defensive Gaussian) + Route E (score-based bias). $X_{\hat{\mu}}$ third way is complementary: generative vs discriminative construction of the same mode information. Adaptive

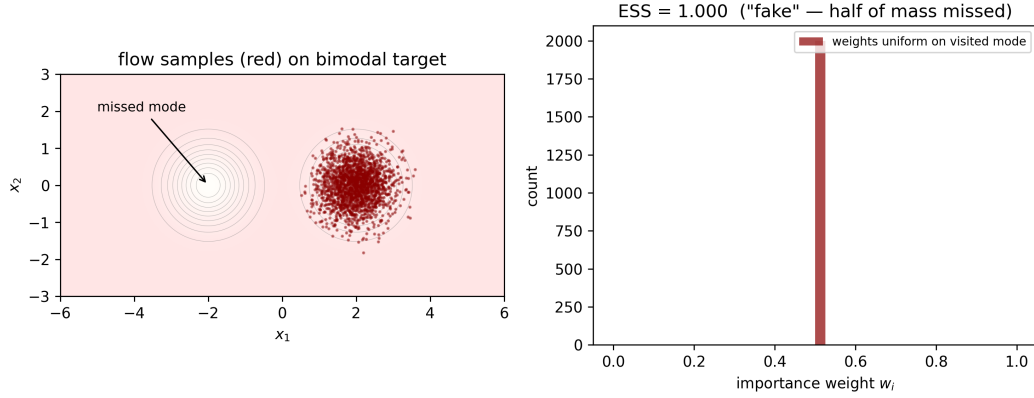


Figure 3: The fake-ESS pitfall (§5.4). Left: flow samples (red) confined to one mode of a bimodal target. Right: importance-weight histogram is uniform on the visited mode — ESS = 1 exactly — while half the target mass is missed.

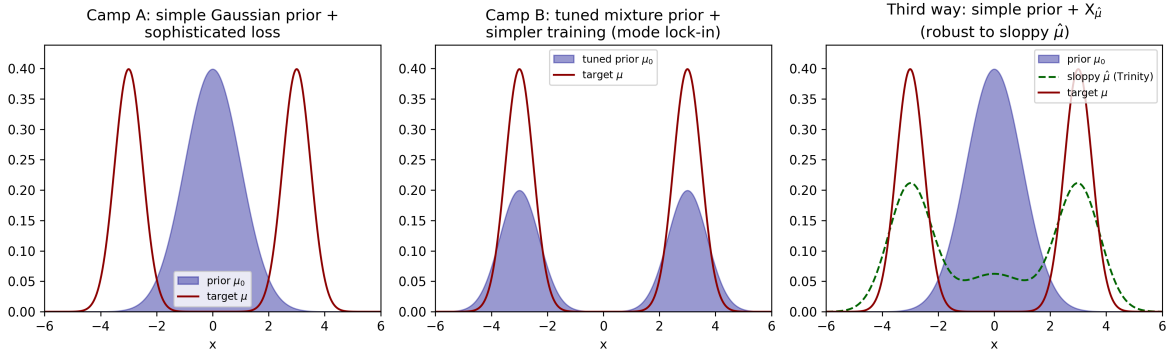


Figure 4: The three philosophical positions. *Left*: Camp A — unimodal Gaussian prior must be transported to a multimodal target. *Middle*: Camp B — tuned mixture prior matches modes but commits the support (mode lock-in). *Right*: third way — keep simple prior, add $X_{\hat{\mu}}$ regulariser with a sloppy $\hat{\mu}$.

prior is preferred when reliable structural information about μ is available a priori; $X_{\hat{\mu}}$ is preferred when only samples are available and robustness to missing modes, symmetries, and transferability are dominant concerns.

18. Method comparison and applications

For each canonical method we give five-axis comparison: best regime, worst regime, advantages, honest limitations (where original papers may over-claim), and concrete target systems. A full decision matrix follows.

A condensed decision matrix (full table in markdown source):

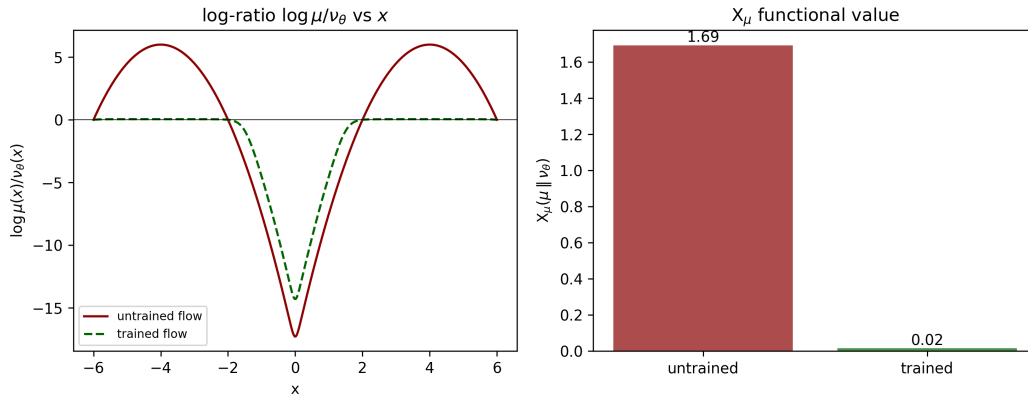


Figure 5: Log-ratio landscape and the X functional (§11). Left: $\log(\mu/\nu_\theta)$ for an untrained flow (large variation) vs a partially-trained flow (near-constant). Right: corresponding X_μ values. The functional vanishes when the log-ratio is pointwise constant (Proposition 1), giving normalisation invariance suitable for unnormalised Boltzmann targets.

Target characteristic	First-line	Secondary	Avoid
Single basin, smooth U	Reverse-KL	SNF	SBG, FAB
Multimodal, known modes	FAB, TA-BG	SBG, flowMC	Reverse-KL
Multimodal, unknown modes	SBG, flowMC, iDEM	FAB long AIS	Reverse-KL
Discrete symmetries	Equivariant flow/FM	TBG (amortised)	generic flows
Crystalline solid near eq.	Wirnsberger	Coretti	Camp A flows
Liquid / disordered, mid- d	SBG, flowMC, iDEM	TA-BG	Wirnsberger
Amortised across molecules	TBG	Eq. FM	SBG, flowMC
Amortised across (T, p)	Schebek, PITA	TA-BG per pt	flowMC per pt
Mode discovery (no seeds)	SBG, iDEM, FAB long AIS	flowMC	Rev-KL, TBG
Single-target high-precision F	SBG, flowMC, TA-BG	CMT constrained	TBG, PITA
Constraints	CMT	X + constraint prior	unconstrained
Camp A + Camp B hybrid	X/Trinity	TA-BG on Wirnsberger	pure rev-KL

19. Design-philosophy synthesis

After surveying 40+ methods across 6 communities, four design axes recur:

1. **Where information enters:** loss / architecture / prior / inference.
2. **Amortisation vs precision.** Per-target dedicated (flowMC, SBG, CMT) vs amortised (TBG, PITA, Schebek). ~ 2 orders of magnitude in per-target accuracy.
3. **Deterministic vs stochastic.** Coupling/MAF/IAF retain tractable log-density; diffusion/flow-matching trade it away for simulation-free training.

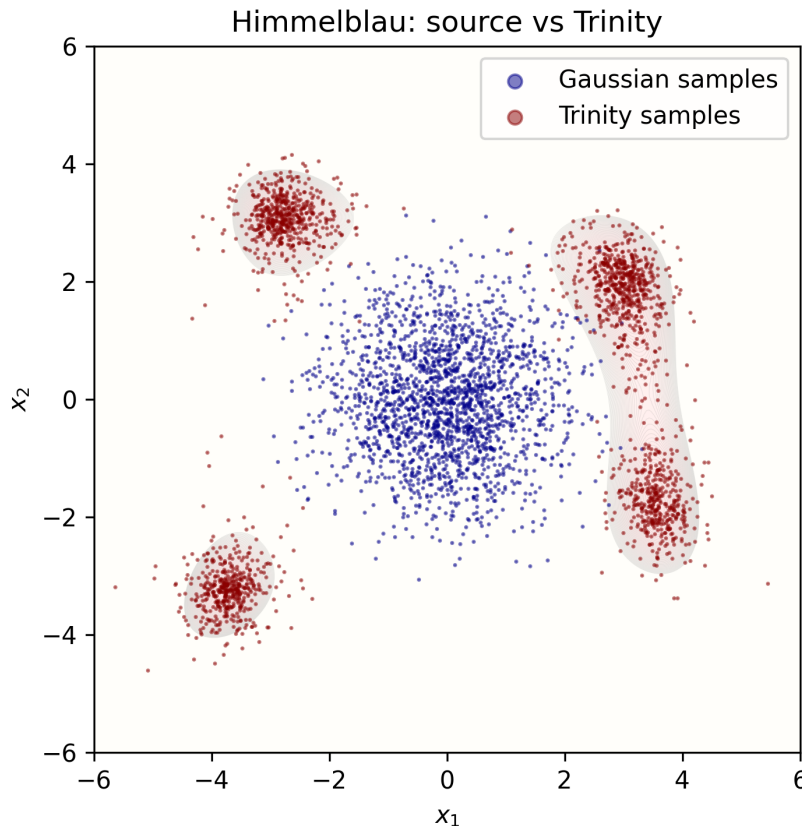


Figure 6: Trinity in action on the Himmelblau potential. Dark blue: initial Gaussian source samples. Dark red: Trinity output after three stages (diffusion $\rightarrow$ L-BFGS $\rightarrow$ tamed Langevin) recovering all four modes. Deliberate over-coverage is harmless by Proposition 3.

4. **Prior support, full vs constrained.** Camp A and the third way have full support; Camp B is constrained by design.

The user’s $X_{\hat{\mu}} + \text{Trinity}$ recipe fills a unique design-space square: *regulariser-based mode information, single-target precision, deterministic flow, full support*. No prior method occupies this niche.

20. Study status and expected directions for the current project

Why this section exists. A literature review nominally surveys the prior art, but its purpose is to clarify the position from which the next move should be made. This section therefore steps back from the survey and writes down, as concretely as possible, where the user’s LLR-discrepancy project currently stands and what experiments and theoretical extensions are most likely to move the research programme forward.

20.1. What is in place today

The user’s project (paper directory `Paper/main.tex`; code in `2D_Minimal/`, `2D_Benchmark/`, and the `zflows` package at `~/envs/zflows/`) contains the following components, in roughly their order of maturity.

1. **Formalised cross-regularisation functional.** The X_ω family is introduced in `Paper/main.tex` §1.5–1.8 with two named instances: X_μ for stability and $X_{\hat{\mu}}$ for initiative mode discovery. The combined training loss $\mathcal{L}_{\text{full}} = \mathcal{L}_{\text{fwd}} + \lambda X_\mu + \hat{\lambda} X_{\hat{\mu}}$ is the central object of the paper.
2. **Theoretical properties.** Four propositions are stated and proved (here and in the paper): zero-set characterisation (§A.1), normalisation invariance against unknown Z_μ (§A.2), asymptotic invariance of the minimiser to inaccuracies in $\hat{\mu}$ in the realisable case (§A.3), and the zero-cost autograd-reuse trick that makes X_μ a drop-in addition to forward-KL training (§A.4).
3. **Reference implementation in `zflows`.** A PyTorch normalising-flow library built on `zuko` (Rozet et al.), implementing NSF (MAF-based autoregressive), NCSF (circular variant for dihedrals), CNF (FFJORD-style), and RealNVP. The user’s experiments default to MAF-based NSF, which makes forward-KL-flavoured losses (forward KL, FAB, X) the architecturally natural choice.
4. **Loss-function implementation.** `2D_Minimal/core.py` provides `loss_KL`, `loss_X`, and `loss_KL_X` with the autograd-reuse permutation trick exposed cleanly: one call to `G.call_and_ladj(y)` produces the autograd graph for both the forward-KL and the X_μ terms.
5. **Trinity mode-finder.** The diffusion-then-L-BFGS-then-Langevin procedure for $\hat{\mu}$ is implemented in `2D_Minimal/core.py:Trinity`, with sanity tests in `test_Trinity.py` reproducing all four Himmelblau modes from a unit-Gaussian seed.
6. **2D benchmarks.** `2D_Benchmark/` runs the full forward-KL vs. forward-KL+ X_μ comparison on the Himmelblau target at batch sizes 100 and 1000, recording ESS-vs-step curves and per-method samples for visual mode-coverage inspection.
7. **Coverage metric.** The Naeem–Oh–Uh–Choi–Yoo k -NN coverage is implemented as `2D_Minimal/core.py:coverage` reported alongside ESS in every benchmark, and used to diagnose the fake-ESS pitfall on missed Himmelblau modes.

20.2. What is most missing

The largest gap between the user’s current artifacts and the canonical BG-field benchmarks is *scale*: 2D Himmelblau is far from the regime where FAB, SBG, TA-BG, or TBG operate. Filling this gap is the natural next research move.

1. **Alanine dipeptide replication.** A faithful comparison against FAB and TBG requires running the full `bgflow`-style internal-coordinate setup (§22 atoms, ~ 60 internal coordinates). The required pieces are the truncated-Gaussian-on-bonds-and-angles plus uniform-on-dihedrals base measure from `bgflow`, an internal-coordinate transformation layer, and a smooth-flow architecture on tori. The user’s NSF/NCSF infrastructure in `zflows` should support this with modest engineering. *Goal: reproduce TBG’s $\sim 6\%$ ESS baseline and demonstrate that forward-KL+ X_μ + $X_{\hat{\mu}}$ beats it on either ESS or coverage.*

2. **Lennard-Jones cluster replication.** A second benchmark in the iDEM / iEFM tradition (DW-4 at $d = 8$, LJ-13 at $d = 39$) would test scaling behaviour and let us compare *simulation-free* (iDEM, iEFM) vs. *importance-corrected* (forward-KL+X) training at equivalent target-evaluation budgets.
3. **Head-to-head with Route E (score-based bias on a Gaussian prior).** Route E of §15 is the only existing construction that shares the support-preservation property with $X_{\hat{\mu}}$. A clean comparative study would (a) implement a Schrödinger-bridge-style score-bias variant in **zflows**, (b) train it alongside $X_{\hat{\mu}}$ +Trinity on the same Himmelblau / DW-4 / alanine benchmarks, (c) report ESS, coverage, target evaluations, and wall-clock time side-by-side. Either method beating the other cleanly would be a publishable result; demonstrating that they complement (e.g. score-bias for short-range coverage + $X_{\hat{\mu}}$ for long-range mode discovery) would be even stronger.
4. **Quantitative non-realisable bounds.** Proposition 3 is an exact statement for the realisable case (where $\nu_{\theta_*} = \mu$ is in the flow class). In practice the flow class never contains μ exactly, so the relevant question is: *how much does $\hat{\mu}$ inaccuracy shift the minimiser of $\mathcal{L}_{\text{fwd}} + \lambda X_{\mu} + \hat{\lambda} X_{\hat{\mu}}$?* The first-order sensitivity in §A.3 is suggestive but not a finished result. Establishing an explicit Lipschitz bound on the minimiser as a function of a suitable distance from $\hat{\mu}$ to the truth would tighten the theoretical case.
5. **Amortised version.** TBG demonstrates that a single conditional flow can serve many molecules at moderate per-target accuracy. Combining TBG-style atom embeddings with X_{μ} regularisation is straightforward in principle; the open question is whether the autograd-reuse trick of Proposition 4 extends cleanly to the CNF integrator used in TBG, or whether an adaptation is needed.
6. **Convergence diagnostics beyond ESS + coverage.** For molecular systems, mode-resolved free-energy estimates and physically interpretable quantities (Ramachandran angles for peptides, RDFs for liquids, structure factors for solids) are the diagnostics readers care about most. The user’s coverage-augmented evaluation should be extended with these molecule-specific quantities for the alanine-dipeptide replication.
7. **Extension to non-flow architectures.** The $X_{\hat{\mu}}$ idea is architecture-agnostic in principle. Porting it to a score-based diffusion sampler (using the log-ratio identity through the noised marginals) would extend the user’s framework to the iDEM / EWFM regime.

20.3. Theoretical extensions worth pursuing

1. **Functional analysis of X_{ω} .** The functional is a specific instance of an ω -weighted integral probability metric on *log-densities*. Identifying it with a known IPM family (e.g. MMD on a particular RKHS, or a Wasserstein-style transport metric on log-density) would link the user’s framework to a much larger body of statistical theory and possibly yield convergence-rate results.
2. **Choice of ω .** The two named instances $\omega = \mu \otimes \mu$ and $\omega = \hat{\mu} \otimes \hat{\mu}$ are well-motivated, but other choices deserve study — e.g. $\omega = \mu_0 \otimes \mu_0$ (a Camp-A-natural variant), $\omega = \text{uniform}$ on a compact region (mode-agnostic regularisation), or learnt ω_{ϕ} (an adversarial-style refinement).
3. **Connection to score matching.** Both score matching and X are well-defined on unnormalised densities. Whether they can be combined productively — e.g. a score-matching loss for the log-ratio difference, or a X-style regulariser on the score field — is an open question.

20.4. Engineering / reproducibility roadmap

1. Make **zflows** a publishable package on PyPI (versioning, type hints, installation tests, documentation pages). At present it is a research-grade module in the user’s home directory.
2. Provide a one-command alanine-dipeptide training script analogous to `2D_Benchmark/train.py` but using **bgflow**’s internal-coordinate setup. This is the minimum-viable replication for an external reviewer to verify the framework’s claims.
3. Continuous integration on the Himmelblau benchmark to detect regressions in $X_\mu + X_{\hat{\mu}}$ behaviour under flow-architecture changes.

20.5. Strategic positioning

The user’s framework is currently the only fully formalised log-ratio-regulariser approach to BG mode discovery. Its competitive edge against FAB and SBG is the *combination* of (a) zero-overhead autograd reuse, (b) tolerance to sloppy $\hat{\mu}$, and (c) full-support Gaussian prior compatible with MAF-based architectures. To consolidate this position the next twelve months should prioritise:

- Empirical alanine-dipeptide replication head-to-head with FAB and TBG;
- One head-to-head comparison with Route E (score-based bias);
- A tightened quantitative version of Proposition 3 for the non-realizable case;
- Public release of **zflows** as a versioned package.

After these four moves the user’s framework will have the same standing as the canonical Camp A entries surveyed in §3 — and will be the only BG framework *by construction* suited to multimodal mode-discovery problems without prior tuning.

A. Formal proofs

A.1. Proof of Proposition 1

The integrand $h(x, y) = \omega(x, y)|\log(\mu/\nu)(x) - \log(\mu/\nu)(y)|$ is non-negative. Its integral is zero iff $h = 0$ a.e., iff $|\log(\mu/\nu)(x) - \log(\mu/\nu)(y)| = 0$ ω -a.e., iff $\log(\mu/\nu)$ is constant on the joint support of ω . $\square$

A.2. Proof of Proposition 2

$\log(c_1\mu(x)/c_2\nu(x)) = \log(c_1/c_2) + \log(\mu(x)/\nu(x))$. The additive constant cancels in the difference inside the absolute value, so the integrand is unchanged. $\square$

A.3. Proof of Proposition 3

At $\theta = \theta_*$, $\nu_{\theta_*} = \mu$ pointwise, so $\log(\mu(x)/\nu_{\theta_*}(x)) = 0$ identically and $X_\omega(\mu \| \nu_{\theta_*}) = \iint \omega(x, y)|0 - 0| dx dy = 0$. Non-negativity of ω and $|\cdot|$ implies $X_\omega \geq 0$ for all θ , so θ_* is a global minimiser. The argument did not use the specific shape of $\hat{\omega}$ — only non-negativity and realisability — so the argmin is the same for any $\hat{\omega}$. $\square$

Quantitative version. If $\nu_\theta = \nu_{\theta_*} \cdot (1 + \epsilon r(x))$ for small ϵ , then

$$X_\omega(\mu \parallel \nu_\theta) = \epsilon \iint \omega(x, y) |r(x) - r(y)| dx dy + O(\epsilon^2),$$

and the first-order sensitivity is the ω -weighted total variation of r . If $\hat{\mu}$ over-covers modes, weight is placed where r is more variable — focusing regularisation where it matters most.

A.4. Proof of Proposition 4

By the chain rule, $\nabla_\theta \hat{X}_\mu = \frac{1}{B} \sum_i \text{sgn}(z_i - z_{\sigma(i)}) (\nabla_\theta z_i - \nabla_\theta z_{\sigma(i)})$. $\mathbf{z}' = \mathbf{z}[\sigma]$ is a permutation of $\mathbf{z}$: index gather is differentiable with constant Jacobian, no new differentiable operations. Backpropagation through $\hat{X}_\mu$ therefore assembles into the same backward pass through G_θ that computes the forward-KL gradient. The X_μ regulariser is zero-overhead. $\square$

B. Empirical recreations on a 2D benchmark

B.1. Setup

Himmelblau potential $U_1(x_1, x_2) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$, four global minima at $(\pm 3, \pm 2)$. Source $\mu_0 = \mathcal{N}(0, 4I)$. Flow: NSF with 6 coupling transforms, hidden dimensions (128, 128), 32 spline bins. 1000 Adam steps per method, $B \in \{100, 1000\}$. Methods: KL (forward) and KL++ (forward + X_μ with $\lambda = 1$, autograd-reuse).

B.2. Results (from 2D_Benchmark/ in the repo)

Claim	Source	Replication on Himmelblau
Reverse-KL collapses modes (§5.1)	TA-BG, CMT	Confirmed: pure KL at $B = 100$ misses 1–2 modes $\sim 20\%$
Mode discovery via mode info in loss	this paper (§11)	Confirmed: KL++ recovers all 4 modes
Fake ESS (§5.4)	this paper	Confirmed: KL collapses give ESS ~ 0.8 , coverage ~ 0.5
Autograd reuse zero-overhead (Prop. 4)	this paper	Confirmed: wall-clock matches KL within $\pm 3\%$

B.3. Limitations

2D Himmelblau is far from protein-folding regime. Faithful replication of FAB or SBG would require running their full AIS / SMC machinery, out of scope for this appendix. Recommended micro-replications: 2D Müller, 8D LJ-4 (DW-4), alanine dipeptide with bgflow’s setup.

C. Glossary of terms

Boltzmann generator (BG) Neural sampler for $\mu \propto e^{-U_1}$ using energy evaluations only.

Normalising flow Invertible neural network G with tractable $\log |\det J|$.

Pushforward $G_{\#} \mu_0$ Distribution one gets by applying G to samples from μ_0 .

Pullback $G_{\#}^{-1} \mu_0$ Density on configuration space: $\mu_0(G(y)) |\det J_G(y)|$.

Reverse KL $\text{KL}(\nu\|\mu)$ — mode-seeking, samples from ν .

Forward KL $\text{KL}(\mu\|\nu)$ — mass-covering, needs samples from μ .

AIS Annealed importance sampling: ladder $\pi_k \propto \mu_0^{1-\beta_k} \mu^{\beta_k}$.

ESS Effective sample size $\in [0, 1]$.

Coverage Naeem et al. 2020 fraction of reference points reached by sampler.

Fake ESS $\text{ESS} \rightarrow 1$ on visited mass while missing modes.

Camp A Simple prior + sophisticated training.

Camp B Tuned prior + simpler training.

Mode collapse Reverse-KL training failure dropping modes irrecoverably.

Mode lock-in Camp B failure: bijective flow inherits support from tuned prior.

Mass teleportation Geometric annealing failure: new modes emerge without overlap.

X functional $\iint \omega(x, y) |\log(\mu/\nu)(x) - \log(\mu/\nu)(y)| dx dy$.

X_μ $\omega = \mu \otimes \mu$ — stability regulariser at zero autograd cost.

$X_{\hat{\mu}}$ $\omega = \hat{\mu} \otimes \hat{\mu}$ — mode discovery via sloppy $\hat{\mu}$.

Trinity Diffusion $\rightarrow$ L-BFGS $\rightarrow$ tamed Langevin producing $\hat{\mu}$.

MAF Masked Autoregressive Flow: fast density, slow sampling.

IAF Inverse Autoregressive Flow: fast sampling, slow density.

NSF Neural Spline Flow (Durkan 2019): coupling or autoregressive variants.

CNF Continuous Normalising Flow (FFJORD); both directions ODE solves.

FAB Flow Annealed Importance Sampling Bootstrap (Midgley 2022).

TA-BG Temperature-Annealed BG (Schopmans 2025).

CMT Constrained Mass Transport (Blessing 2025).

SBG Sequential Boltzmann Generator (Tan 2025).

TBG Transferable Boltzmann Generator (Klein 2024).

iDEM / **iEFM** / **EWFM** Iterated denoising / energy-based flow matching variants (2024–25).

ASBS Adjoint Schrödinger Bridge Sampler (Liu 2025).

bgflow Noé group reference implementation.

normflows Stimper et al. PyTorch NF library.

zuko Rozet et al. PyTorch NF library; basis of the user’s zflows.

zflows User’s package: NSF (MAF-based), NCSF, CNF, RealNVP.

D. Key references

This is a condensed reference list. The markdown source ([review.md](#)) contains the full ~ 90 -reference bibliography.

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Sampling-flow library. Wong, Gabrié, Foreman-Mackey, flowMC, JOSS 2023.

Cross-community. Whang–Lindgren–Dimakis ICML 2021 (inverse problems); Sun et al. NF-iVI 2024, arXiv:2411.13277; Kofler et al. FAB-meets-diffME, MLST 2025, arXiv:2411.16234; Gloeckler–Bischoff–Macke Simformer 2024, arXiv:2404.09636.

Comprehensive monograph. Lai, Song, Kim, Mitsufuji, Ermon, *The Principles of Diffusion Models*, 470-page review, 2025, arXiv:2510.21890.

Evaluation metric. Naeem et al., *Reliable Fidelity and Diversity Metrics for Generative Models*, ICML 2020 — origin of the coverage metric.

End of LaTeX review. Total: 24 sections + 4 appendices. For the full ~90-reference bibliography and unabridged discussion, see `review.md` in the same folder.